

The platform has been assembled and is ready to collect data. In the current stage, the sensor platform is capable of reading temperature, humidity, and the levels of carbon monoxide, tropospheric ozone, and combustion gas. All gases are measured in parts per million (ppm). There will be a total of five air quality sensor boards that will communicate to a server to deposit the data. The developer is still working on the visualisations, after which the system will be ready for deployment.

## Making the Khan Academy more accessible

<http://pi.mujica.org/>

The Khan Academy is becoming one of the most acclaimed online content libraries, with many students trusting its lessons more than they trust their teacher. The simple and easy to understand lessons with graphic representations of almost all concepts spanning fields from physics to business are indeed a boon even for adults who want to learn more. If you ever felt that the Khan Academy would be more useful in remote areas than in urban centres, you'll be happy to hear of KA-Pi, a project that aims to take Khan Academy content to students who have no Internet connectivity.

**The open twist:** KA-Pi is a simple, plug-and-play server solution to play the Khan Academy videos where no Internet access is available. By using a green, low-power, small Raspberry Pi computer, the solution becomes all the more accessible. You just need to



download the KA-Pi card image, burn it to a 16 GB SD card and plug the card into your Raspberry Pi. The card image includes the operating system, a Web server and lightweight content, making it a truly plug and play system. The content can even be streamed to a local network. The content, from 'Khan Academy on a Stick', comprises simple MP4 encoded videos that will play on browsers, iPads, iPhones and Android devices without the use of Flash. 'Khan Academy on a Stick' content is open sourced using a Creative Commons licence.



## Ensuring local, clean and sustainable food production

<http://hapih.com/>

One often wonders how technology can help solve social and economic problems, and one of the biggest such problems is that of the availability of food. Can technology, especially open source technology, help improve food production and ensure better nutrition for all? The Hydroponic Automation Platform Initiative (HAPI) led by Tyler Reed with sponsorship from the Human Services Research and Technology Institute, Washington, is an attempt towards this end. Hydroponic irrigation is a way of growing food using the minerals in water, without soil. It is a lesser known technique, but definitely easier to adopt in urban areas. So, by providing a complete connected platform with automation modules, structural designs, a clean seed network and a best practice application, HAPI hopes to promote urban hydroponics, thereby improving collective yield and ensuring sustainable food production.

**The open twist:** HAPI is the first worldwide open source initiative for developing scalable hydroponic and aquaponic structures and automation systems (<http://sourceforge.net/p/hydroplatform/wiki/Home/>). The project is still in the fund-raising and planning stage. There is, however, a spot of work going on concerning individual components and modules. After zeroing in on the hardware modules and management software functions, the team will be looking at creating firmware to control lighting, feeding, pH levels, TDS/eC levels and possibly complete reservoir refreshes. They will be exploring connectivity between HAPI-based systems and the outside world, e.g., system control via ssh or the Web. The project is seeking urban collaborators to develop rooftop hydroponic farms, as well as developers familiar with Arduino and Raspberry Pi. They will be raising funds on Kickstarter. 

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